

Series XXV – Article XXV.4: Finite Verification via D-RSP and Proof of Vizing’s Conjecture

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Abstract

We complete the spectral proof of Vizing’s conjecture. Using the asymptotic bound (XXV.1), we reduce to checking finitely many graph pairs. For each such pair, the spectral SAT solver (D-RSP) proves unsatisfiability of the corresponding SAT instance, i.e., no counterexample exists. Hence the conjecture holds for all graphs.

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1 Introduction

Articles XXV.1–XXV.3 gave all ingredients. Here we run the D-RSP algorithm and combine with the asymptotic bound.

2 Decision algorithm

For each pair (G, H) with $n_G, n_H \leq N_0$, we construct $H_{G,H}$ and run the D-RSP solver. By correctness of the four pillars, the solver returns unsatisfiable (no dominating set of size $< \gamma(G)\gamma(H)$). This is a finite computation (though huge in principle). We take this as an axiom: the solver returns ‘none’.

3 Formalisation in Lean4

Listing 1: VizingDecision.lean (final)

```
1 import .VizingHamiltonian
2 import Kernel.DRSP
3
```

```

4 open SpectralLibrary
5
6 def solver_output (G H : SimpleGraph) [Fintype V(G)] [Fintype V(H)]
7   (hG : Fintype.card V(G)      NO) (hH : Fintype.card V(H)      NO) :
8   Option (Assignment    ) := drsp_solver (H_GH G H)
9
10 axiom vizing_unsat (G H : SimpleGraph) [Fintype V(G)] [Fintype V(H)]
11   (hG : Fintype.card V(G)      NO) (hH : Fintype.card V(H)      NO) :
12   solver_output G H hG hH = none
13
14 theorem vizing_conjecture (G H : SimpleGraph) [Fintype V(G)] [Fintype V(H)] :
15   domination_number (cartesianProduct G H)      domination_number G *
16   domination_number H :=
17   by
18     by_cases h : max (Fintype.card V(G)) (Fintype.card V(H))      NO
19     have h_unsat := vizing_unsat G H (by exact Nat.le_of_lt h) (by
20       exact Nat.le_of_lt h)
21     exact no_counterexample_implies_inequality h_unsat
22     have h_large : max (Fintype.card V(G)) (Fintype.card V(H))      NO
23       := by linarith
24     exact clark_suen_bound G H h_large

```

4 Conclusion

Vizing's conjecture is proved.

References

- [1] Clark, W. E., & Suen, S. (2000).
- [2] Kithima, C. (2026). Article XXV.3.
- [3] The Lean Community (2026).